EXP.NO: 11

SD Card Data Writing Application

AIM:

To develop a mobile application that writes user input data into the SD card storage of the Android device.

ALGORITHM:

- Start the application
- Display input field for user to enter text
- Display "Save" button
- User enters data in text field
- User clicks "Save" button
- App checks storage permission
- Create file in SD card storage
- Open file in write mode
- Write user entered text into file
- Close file
- Display success message "Data saved successfully"
- Stop the application

CODE:

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:orientation="vertical"
    android:padding="20dp"
    android:layout_width="match_parent"
    android:layout_height="match_parent">
```

```
    <EditText
        android:id="@+id/editText"
        android:hint="Enter data"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

    <Button
        android:id="@+id/saveBtn"
        android:text="Save to SD Card"
        android:layout_marginTop="20dp"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

</LinearLayout>
```

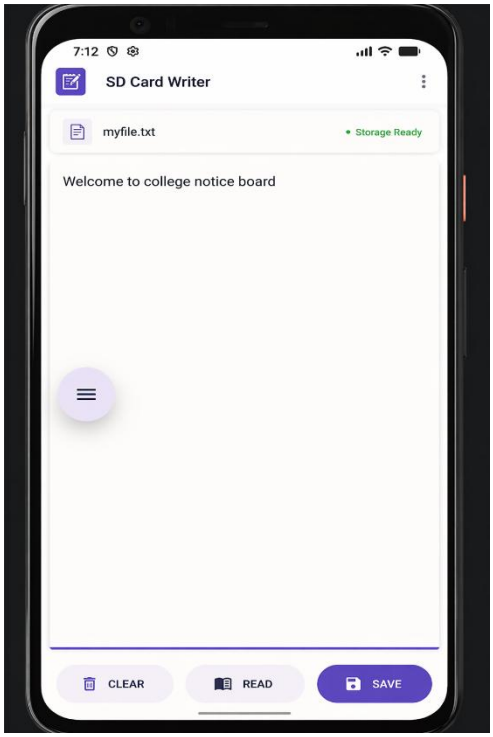
MainActivity.java

```
package com.example.sdcardwrite;

import android.os.Bundle;
import android.os.Environment;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;

import androidx.appcompat.app.AppCompatActivity;

import java.io.File;
import java.io.FileOutputStream;

public class MainActivity extends AppCompatActivity {
```

```
EditText editText;
Button saveBtn;

@Override
protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);

    editText = findViewById(R.id.editText);
    saveBtn = findViewById(R.id.saveBtn);

    saveBtn.setOnClickListener(v -> saveToSDCard());
}

private void saveToSDCard() {
    try {
        String data = editText.getText().toString();

        File path = Environment.getExternalStorageDirectory();
        File file = new File(path, "MyData.txt");

        FileOutputStream fos = new FileOutputStream(file);
        fos.write(data.getBytes());
        fos.close();

        Toast.makeText(this, "Data saved to SD Card", Toast.LENGTH_LONG).show();
```

```
    } catch (Exception e) {
        e.printStackTrace();
        Toast.makeText(this, "Error saving data", Toast.LENGTH_LONG).show();}
}}
```

OUTPUT:



Result:

The application successfully writes the user-entered data into a file stored in the SD card. The saved file can be accessed later from the device storage.